09/10/26

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Genetic optimization algorithm :

Step1 : Start
Step2 : Initialize a population of chromosomes randomly, representing possible solutions to the optimization problem.
Step3 : Evaluate the fitness of each chromosome using the fitness function
Step4 : Check whether the termination condition is satisfied (maximum generation reached or optimal solution found). If yes, go to step 9
Step5 : Select parent chromosomes based on their fitness values.
Step6 : Apply coorsover to the selected parents to generate new offspring
Step7 : Apply mutation to the offspring to introduce genetic variation
Step8 : Form a new population using the offspring and, if required, the best individuals from the previous population. Go to step 3
Step9 : Output the best chromosome as the optimal or best-found solution
Step10 : Stop.

Renewable energy optimization : solar, wind, battery
improve utilization → energy cost minimize

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Hour	Solar	Wind	Demand	Battery Power
16	3.0	4.0	8.0	1.07
17	1.0	4.0	8.0	4.43
18	0.0	5.0	8.0	4.25
19	0.0	6.0	8.0	1.93
20	0.0	5.0	8.0	4.32
21	0.0	4.0	8.0	1.67
22	0.0	3.0	8.0	4.95
23	0.0	3.0	8.0	2.14
~~24~~	~~0.0~~			
~~25~~	~~0.0~~			

b) optimal wind energy allocation
c) optimal battery charging/discharging schedule
d) Minimum estimated energy cost
e) Renewal energy utilization
7) Stop

O/P: RENEWABLE ENERGY OPTIMIZATION RESULTS

Total Energy Cost : 124.52
Renewable Energy Utilization : 72.92%
Grid Energy Required : 23.50 KWh
Optimization Fitness Value : 123.71

HOURLY BATTERY SCHEDULE

Hour	Solar	Wind	Demand	Battery Power
0	0.0	3.0	8.0	1.22
1	0.0	3.0	8.0	3.29
2	0.0	4.0	8.0	-1.91
3	0.0	4.0	8.0	-2.14
4	0.0	5.0	8.0	3.62
5	1.0	4.0	8.0	-0.07
6	3.0	3.0	8.0	-4.38
7	5.0	3.0	8.0	-0.87
8	7.0	4.0	8.0	-4.90
9	9.0	5.0	8.0	-4.85
10	10.0	4.0	8.0	-4.23
11	11.0	3.0	8.0	-4.73
12	10.0	4.0	8.0	-4.89
13	9.0	5.0	8.0	2.74
14	7.0	6.0	8.0	4.11
15	5.0	5.0	8.0	1.25

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Renewable Energy optimization Using Genetic Algorithm:

Problem Statement: the objective of this application is to minimize energy costs and maximize renewable energy utilization using solar, wind and battery storage.
A Genetic Algorithm is used to determine the optimal energy allocation, battery charging and discharging schedules, and reduce dependence on grid electricity while meeting energy demand efficiently

Algorithm:

1) Start
2) Initialize population randomly representing solar, wind, and battery schedules.
3) Evaluate the fitness of each individual based on:
a) Total energy cost
b) Renewable energy utilization
4) While termination condition is not satisfied do
   a) select parent solutions based on fitness
   b) apply crossover to generate new solutions
   c) apply mutation to introduce variation
   d) evaluate the fitness of the new solutions
   e) select best solution to form the next generation
5) Identify the best solution that minimizes total energy cost and maximizes renewal energy utilization while meeting demand
6) output:
   a) optimal solar energy allocation

1BM24CS104_RENEWABLE ENERGY OPTIMIZATION.ipynb

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Commands + Code + Text Run all

[2] ✓ 1s

```
import numpy as np
import random

# ----------------------------
# 1. INPUT DATA (24 HOURS)
# ----------------------------
hours = 24

solar = np.array([
    0, 0, 0, 0, 0, 1, 3, 5, 7, 9, 10, 11,
    10, 9, 7, 5, 3, 1, 0, 0, 0, 0, 0, 0
], dtype=float)

wind = np.array([
    3, 3, 4, 4, 5, 4, 3, 3, 4, 5, 4, 3,
    4, 5, 6, 5, 4, 4, 5, 6, 5, 4, 3, 3
], dtype=float)

demand = np.full(hours, 8.0) # kW

price = np.array(
    [5]*6 + [7]*6 + [9]*6 + [6]*6,
    dtype=float
) # Cost per kWh (illustrative)

BATTERY_CAPACITY = 20.0 # kWh
MAX_BATTERY_POWER = 5.0 # kW
EFFICIENCY = 0.95

POPULATION_SIZE = 50
GENERATIONS = 100
MUTATION_RATE = 0.1

# ----------------------------
# 2. FITNESS FUNCTION
# ----------------------------
def evaluate(chromosome):
    soc = BATTERY_CAPACITY / 2
    total_cost = 0.0
    renewable_used = 0.0
    grid_energy = 0.0
    unmet_demand = 0.0
```

1BM24CS104_RENEWABLE ENERGY OPTIMIZATION.ipynb

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[2] ✓ 1s

```
        # Grid provides any remaining demand
        grid = max(0.0, remaining_demand)
        grid_energy += grid
        total_cost += grid * price[t]
        unmet_demand += max(0.0, remaining_demand - grid)

    utilization = renewable_used / max(np.sum(demand), 1) * 100

    # Minimize cost and penalize renewable curtailment.
    # A reward encourages higher direct renewable utilization.
    objective = (
        total_cost
        + 0.1 * curtailed_energy
        - 0.01 * renewable_used
        + 1000 * unmet_demand
    )

    return objective, total_cost, utilization, grid_energy


# ----------------------------
# 3. INITIAL POPULATION
# ----------------------------
def create_chromosome():
    return np.random.uniform(
        -MAX_BATTERY_POWER,
        MAX_BATTERY_POWER,
        hours
    )


# ----------------------------
# 4. GENETIC ALGORITHM
# ----------------------------
def genetic_algorithm():
    population = [
        create_chromosome()
        for _ in range(POPULATION_SIZE)
    ]

    for generation in range(GENERATIONS):
        population.sort(key=lambda x: evaluate(x)[0])
```

1BM24CS104_RENEWABLE ENERGY OPTIMIZATION.ipynb

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Commands + Code + Text Run all

[2] ✓ 1s

```
for t in range(hours):
    print(
        f"{t:4} | {solar[t]:5.1f} | {wind[t]:4.1f} | "
        f"{demand[t]:6.1f} | {best_schedule[t]:13.2f}"
    )

print("-" * 55)
```

```
RENEWABLE ENERGY OPTIMIZATION RESULTS
==========================================
Total Energy Cost: 124.52
Renewable Energy Utilization: 72.92%
Grid Energy Required: 23.50 kWh
Optimization Fitness Value: 123.71

HOURLY BATTERY SCHEDULE
-----------------------------------------------------
Hour | Solar | Wind | Demand | Battery Power
-----------------------------------------------------
   0 |   0.0 |  3.0 |    8.0 |          1.22
   1 |   0.0 |  3.0 |    8.0 |          3.29
   2 |   0.0 |  4.0 |    8.0 |         -1.91
   3 |   0.0 |  4.0 |    8.0 |         -2.14
   4 |   0.0 |  5.0 |    8.0 |          3.62
   5 |   1.0 |  4.0 |    8.0 |         -0.07
   6 |   3.0 |  3.0 |    8.0 |          4.38
   7 |   5.0 |  3.0 |    8.0 |         -0.87
   8 |   7.0 |  4.0 |    8.0 |         -4.90
   9 |   9.0 |  5.0 |    8.0 |         -4.85
  10 |  10.0 |  4.0 |    8.0 |         -4.23
  11 |  11.0 |  3.0 |    8.0 |         -4.73
  12 |  10.0 |  4.0 |    8.0 |         -4.89
  13 |   9.0 |  5.0 |    8.0 |          2.74
  14 |   7.0 |  6.0 |    8.0 |          4.11
  15 |   5.0 |  5.0 |    8.0 |          1.25
  16 |   3.0 |  4.0 |    8.0 |          1.07
  17 |   1.0 |  4.0 |    8.0 |          4.43
  18 |   0.0 |  5.0 |    8.0 |          4.25
  19 |   0.0 |  6.0 |    8.0 |          1.93
  20 |   0.0 |  5.0 |    8.0 |          4.32
  21 |   0.0 |  4.0 |    8.0 |          1.67
  22 |   0.0 |  3.0 |    8.0 |          4.95
  23 |   0.0 |  3.0 |    8.0 |          2.14
-----------------------------------------------------
```

1BM24CS104_RENEWABLE ENERGY OPTIMIZATION.ipynb

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Commands + Code + Text Run all

[2] ✓ 1s

```
best_schedule = genetic_algorithm()

objective, cost, utilization, grid_energy = evaluate(best_schedule)


# ----------------------------
# 6. DISPLAY RESULTS
# ----------------------------
print("\nRENEWABLE ENERGY OPTIMIZATION RESULTS")
print("=" * 45)

print(f"Total Energy Cost: {cost:.2f}")
print(f"Renewable Energy Utilization: {utilization:.2f}%")
print(f"Grid Energy Required: {grid_energy:.2f} kWh")
print(f"Optimization Fitness Value: {objective:.2f}")

print("\nHOURLY BATTERY SCHEDULE")
print("-" * 55)
print("Hour | Solar | Wind | Demand | Battery Power")
print("-" * 55)

for t in range(hours):
    print(
        f"{t:4} | {solar[t]:5.1f} | {wind[t]:4.1f} | "
        f"{demand[t]:6.1f} | {best_schedule[t]:13.2f}"
    )

print("-" * 55)
```

```
RENEWABLE ENERGY OPTIMIZATION RESULTS
=============================================
Total Energy Cost: 124.52
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HOURLY BATTERY SCHEDULE
-------------------------------------------------------
Hour | Solar | Wind | Demand | Battery Power
-------------------------------------------------------
   0 |   0.0 |  3.0 |    8.0 |          1.22
   1 |   0.0 |  3.0 |    8.0 |          3.29
   2 |   0.0 |  4.0 |    8.0 |         -1.91
```

[2] ✓ 1s

```
            population.sort(key=lambda x: evaluate(x)[0])

            # Preserve the best 10 solutions
            new_population = population[:10]

            while len(new_population) < POPULATION_SIZE:
                parent1, parent2 = random.sample(population[:25], 2)

                # Crossover
                point = random.randint(1, hours - 1)
                child = np.concatenate(
                    (parent1[:point], parent2[point:])
                )

                # Mutation
                for i in range(hours):
                    if random.random() < MUTATION_RATE:
                        child[i] = random.uniform(
                            -MAX_BATTERY_POWER,
                            MAX_BATTERY_POWER
                        )

                new_population.append(child)

            population = new_population

        return min(population, key=lambda x: evaluate(x)[0])


    # ----------------------------
    # 5. RUN OPTIMIZATION
    # ----------------------------
    best_schedule = genetic_algorithm()

    objective, cost, utilization, grid_energy = evaluate(best_schedule)


    # ----------------------------
    # 6. DISPLAY RESULTS
    # ----------------------------
    print("\nRENEWABLE ENERGY OPTIMIZATION RESULTS")
    print("=" * 45)

    print(f"Total Energy Cost: {cost:.2f}")
```